



Department of Electronic and Telecommunication

Engineering

(Biomedical Engineering)

University of Moratuwa



Team VitaSync

System for Detecting and Monitoring Respiratory System Related Diseases in Infants and Elders

Project Proposal



Engineering Design Project – BM1190

Name	Index Number
Thennakoon U.I.D.	240657J
Pitiduwa S.M.	240518K
Tennekone A.D.T.M.S.H.	240642J
Wanninayaka W.M.I.R.	240695X

Problem Statement

Chronic Respiratory Diseases (CRDs) account for 12% of deaths in South Asia¹. While early detection significantly improves outcomes, a critical gap exists in accessible, low-cost diagnostic and monitoring technology. Current clinical tools are often centralized in urban hospitals, leading to the neglect of early symptoms in rural and low-income populations and resulting in preventable mortality.

Justification of Problem Selection

Selection of Chronic Respiratory Diseases (CRD) such as Chronic Obstructive Pulmonary Disease (COPD), Asthma and Sleep Apnea management in South Asia as a design focus is justified by the critical diagnostic gap that currently contributes to 12% of regional mortality¹. While these diseases are manageable if detected early, the current medical infrastructure relies heavily on centralized, high-cost diagnostic tools such as hospital grade spirometers that are inaccessible to the rural majority. This creates a technical failure where early signs are neglected not just due to a lack of awareness, but due to a lack of affordable, point-of-care monitoring technology. From an engineering perspective, there is a clear demand for a portable and affordable solution that can provide consistent, non-invasive data logging. By automating the monitoring process, a design intervention can shift the healthcare model from “reactive treatment” of the worst possible outcomes to “proactive management” during early stages, reducing the disease burden and associated productivity loss.

Proposed Solution

A wearable wrist band and chest strap powered by battery (Li-Po battery) that measures SpO₂ and heart rate using a high sensitivity pulse oximetry and heart-rate module along with respiratory rate/breathing effort using an accelerometer and temperature using a temperature sensing module. Different respiratory diseases such as Asthma, COPD and sleep apnea show different levels of drops in Oxygen. Respiratory rate is tracked by measuring the physical expansion and contraction of the chest or neck. Raw data is collected using these sensors using a relatively low cost ESP32 or Arduino-based system. A Machine Learning model is then implemented to analyze the data, detect trends and alert the necessary personnel via an emergency SMS alert and real time visual and audio warnings. System should be implemented using low cost hardware and compact ensuring ease of use.

References

[1] World Health Organization - Chronic respiratory diseases in WHO South-East Asia – World Health Organization[Online] - [Chronic respiratory diseases in WHO South-East Asia.pdf](#)